

Final Thesis - Solar Flares vs Earthquakes

Henrik B. Sørensen

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Introduction

This is my final project of the Data Science: Capstone, HarvardX PH125.9x. A course in Machine Learning based on the R programming language. In this project, I will take an unbiased look into a controversial theory put forward in a YouTube video, I watched years ago: Ben Davidson: Disaster Prediction - <https://www.youtube.com/watch?v=RSfsGTAKp78>.

A similar claim has recently been put out there LiveScience: Solar flares may be triggering earthquakes, controversial study claims - <https://www.livescience.com/planet-earth/solar-flares-may-be-triggering-earthquakes-controversial-study-claims>.

In this project, I will take an unbiased look at the data available to see, if there's an obvious correlation between the solar flare emissions and subsequent Earthquakes. I have no stake in the original theory (or derived theories). I have no affiliation with any of the people involved in either. Nor do I have any personal and/or monetary interest in proving the theory correct. I'm just a curious person by nature and go where the data takes me.

Originally, I wanted to use Machine Learning to predict system failures in the build farm at work (doing approximately 13,000,000 builds/month). However, due to legal reasons (and the lead time of the legal department), the log files of the systems could not be made publicly available. This project will use the same methodology as analyzing these logs: a data trend can lead to an issue with a time delay. And the techniques are approximately the same: a "window" moving over time trying to find a relation between data and result.

I'm an Engineer and see the world as a mix of systems and data: Anything can be modeled as a system ingesting data/influence and producing a well defined and well predicted output. The challenge is the actual model itself.

Plan of attack

- Setup the environment
 - Install necessary libraries (if missing)
 - Include necessary libraries
- Prepare the data set for initial analysis
 - Download data set (if missing) and unpack
 - Split the data set into training set / test set
- Analyze the data set
 - Look into data structures
 - Visualize data to understand distributions and patterns
 - Identify any patterns
- Model Development
 - Step 1: Identify solar flare "profile"
 - Step 2: Attempt to identify a "lag" of input vs output
 - Step 3: Compare any findings with actual results
- Model Testing and Verification
- Model Analysis
- Conclusion

Tools being used

Tool	Version	Usage
RStudio	RStudio 2026.07.1+147 Pacific Dogwood for windows	Main IDE for Developing and running the code
Claude	Claude for Windows, Version 1.24012.9	Assistant for validating syntax and web search
Notepad++	Notepad++ 8.9.1	Scratch Pad while editing R/Rmd code
PowerShell	PowerShell 7.6.3	Various scripting

Data sources

Source	Url
USGS Earthquake Catalog	https://earthquake.usgs.gov/fdsnws/event/1/
Geomagnetic Kp index	https://kp.gfz.de/app/files/Kp_ap_Ap_SN_F107_since_1932.txt
SILSO daily sunspot number	https://www.sidc.be/SILSO/DATA/SN_d_tot_V2.0.csv
NOAA solar flare event lists	https://www.ncei.noaa.gov/products/space-weather/legacy-data/solar-flares-events

Disclaimer!!!!!!

I'm not a Geophysicist nor an Astrophysicist!!! If it turns out, that there's a correlation between the solar flares data and the earthquake data, I will not attempt to explain the actual cause of this, the mechanics behind it or attempt to come up with an actual solution to attempt to predict earthquakes. If it turns out, that I cannot find any correlation, I will not dismiss any of the work of any physicist.

Short Physics Introduction

Just to have a basic understanding of what earthquakes and solar flares are, let's briefly touch upon what, these are:

Earthquakes

Earthquakes are defined as a sudden shaking of the surface of the Earth. This can be caused by erupting volcanoes, mining, explosions and moving of the crust of the Earth (including tectonic plates). The result of these can be shaking of the ground so powerful, that it causes buildings to collapse. It can also cause tsunamis and landslides. Earthquakes are measured on the (Open) Richter Scale. This scale is logarithmic. Meaning, that a 2 is 10x more powerful as a 1. More can be read about Earthquakes at Britannica - Earthquakes - <https://www.britannica.com/science/earthquake-geology/Intensity-and-magnitude-of-earthquakes>

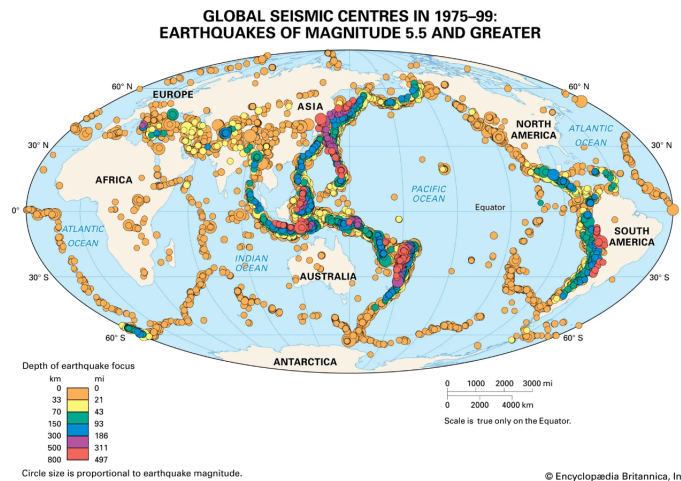


Figure 1: Epi centers of earthquakes

Solar Flares

Solar flares are “explosions” on the surface of the Sun sending out energy, high speed particles and light. This energy surge can overload power grids on the Earth as well as damage electronics. Predicting solar flares could be useful in order to protect power grids. Solar eruptions are measured on a logarithmic scale and labelled A, B, C, M and X. More can be read about solar flares at NASA- What are solar flares - <https://science.nasa.gov/science-research/heliophysics/space-weather/solar-flares/what-is-a-solar-flare/>

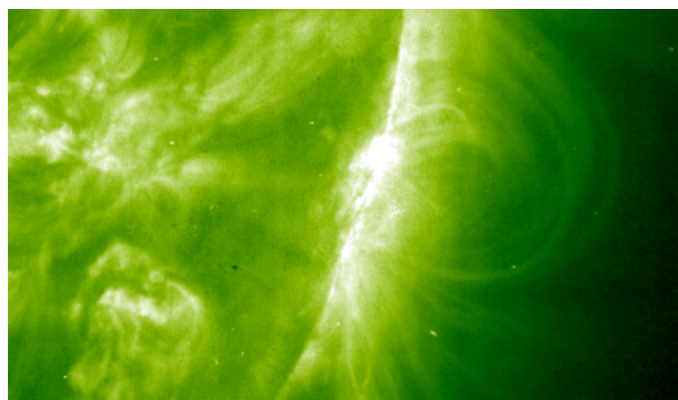


Figure 2: Solar flare, NASA

Environment setup

Environment setup

To analyse the data and perform all the analysis, I intend to perform on the data itself, some packages need to be installed:

```
# Create a list of required packages
# here      - assisting location file system
# httr      - handling HTTP requests (GET/POST) for retrieving data
# readr     - reading csv files
# dplyr     - working with data frames in memory
# tidyr     - generally working with structured data
# lubridate - handling times and dates
# purrr     - handling functional programming
# slider    - working with rolling window
# ggplot2   - handling plots
# car       - supporting further analysis and for testing the model
# lmtest    - checking the model and assumptions
# tinytex   - supporting the formatting of the resulting report
# ncd4      - parsing Net CDF files
required_packages <- c("here", "httr", "readr", "dplyr", "tidyr", "lubridate", "purrr", "slider",
                      "ggplot2", "car", "lmtest", "tinytex", "ncdf4")

# Set default URLs to point to CRAN Mirror (to simplify installing packages)
if (length(getOption("repos")) == 0 || getOption("repos")["CRAN"] == "@CRAN@")
  options(repos = c(CRAN = "https://cloud.r-project.org"))

# Discover missing packages
missing_packages <- required_packages[!required_packages %in% installed.packages()[, "Package"]]

# Install any missing packages
if (length(missing_packages))
  install.packages(missing_packages)

# Load libraries hiding the output from the results
invisible(lapply(required_packages, library, character.only = TRUE))
```

Data retrieval

The different data sources have different ways of retrieving data:

Source	Url	Method
USGS Earthquake Catalog	https://earthquake.usgs.gov/	GET
Geomagnetic Kp index	https://kp.gfz.de/	Download
SILSO daily sunspot number	https://www.sidc.be/SILSO/	Download
NOAA solar flare event lists	https://www.ncei.noaa.gov/	Download Recursive

This means, that we must use different ways of getting the data based on the URL. Basically, we need 3 different functions for handling file download:

```
# Force the data folder to be in location relative to this file by setting the
# work directory to current location
dir_current_location <- here::here()
#setwd(dir_current_location)

# Create function for downloading file from url into destination
```

```

# Only download file if this is missing from the destination
download_file <- function(url, destination_file) {

# download.file() does not create missing directories itself - it just
# errors out, so the destination folder (e.g. "data/" next to this .Rmd)
# must be created first, before checking whether the file already exists.
  dir.create(dirname(destination_file), recursive = TRUE, showWarnings = FALSE)

  if (!file.exists(destination_file))
    download.file(url, destination_file, mode = "wb")
}

```

This will work for the “Geomagnetic Kp index” and “SILSO daily sunspot number”, which are text files. Since, these file are rather small, we rely on the presence of them to determine whether or not to download them.

```

# Let's download the 2 text files to a data folder
# Only download file if this is missing from the destination
geomag_kp_gfz_url <- "https://kp.gfz.de/app/files/Kp_ap_Ap_SN_F107_since_1932.txt"
geomag_kp_gfz_data_file <- "data/geomag_kp_gfz.txt"
sunspot_silso_url <- "https://www.sidc.be/SILSO/DATA/SN_d_tot_V2.0.csv"
sunspot_silso_data_file <- "data/sunspot_silso.csv"

# Download both files
download_file(geomag_kp_gfz_url, geomag_kp_gfz_data_file)
download_file(sunspot_silso_url, sunspot_silso_data_file)

```

The NOAA data is located inside a file structure as shown in the following image:



<u>Name</u>	<u>Last modified</u>	<u>Size</u>
Parent Directory	-	-
1974/	2020-01-24 00:48	-
1975/	2020-01-24 00:48	-
1976/	2020-01-24 00:48	-
1977/	2020-01-24 00:48	-
1978/	2018-07-16 20:50	-

Figure 3: Solar flare, NOAA

This means, that we need to traverse this structure to download all data (.csv files). First, we make a generic function for downloading multiple files at a time.

```

# Create function for downloading files from url into destination folder
# Only download file if this is missing from the destination
#
# This, in the future, could be optimized to load each "root" folder and parse
# the received HTML response. But, this is outside the scope for now.
download_files <- function(urls, destination_folder = "~/data") {

  # Get actual full path for the destination folder
  destination_folder <- path.expand(destination_folder)

  # Loop through the provided URLs
  for (url in urls) {
    # Extract the path of the destination relative for the root destination
    # using Regular expressions

```

```

# Visit https://www.regexg.com/regex-quickstart.php for more information
# about Regular Expressions
relative_path <- sub("^https?:[/\\]+/", "", url)
destination_file <- file.path(destination_folder, relative_path)

# Attempt to query the Url.
# If the status code returned is 200 => SUCCESS! The file can be downloaded.
# Otherwise, skip the file
ok <- tryCatch(status_code(HEAD(url)) == 200, error = function(e) FALSE)
if (!ok) next

# Use the previous function to download the file while creating the
# destination folder is missing.
download_file(url, destination_file)
}
}

```

The NOAA archive has (at least) two different file layouts, so building the list of candidate URLs is specific to each layout - only the downloading itself is shared. In order to ease the download of the entire file system, we use a “hit-and-miss” strategy. Attempting to download data based on a pattern and not the actual data on the server. In future optimizations, an HTML parser COULD be added to facilitate just downloading the available data.

```

# Legacy GOES 1-15 archive: one CSV per year/month/satellite, e.g.
# .../access/avg/2018/07/goes15/csv/g15_xrs_1m_20180701_20180731.csv
# Rather than guess which satellite covered a given month, every candidate
# is included and download_files_recursively() will skip the ones that returns
# a 404 error.
goes_legacy_xrs_urls <- function(url_pattern,
                                years = 1975:as.integer(format(Sys.Date(),
                                                                "%Y")),
                                satellites = sprintf("goes%02d",
                                                    c(15, 14, 13, 12, 11, 10, 9, 8))) {

  urls <- character(0)
  for (year in years) {
    for (month in 1:12) {
      month_str <- sprintf("%02d", month)
      month_start <- as.Date(sprintf("%d-%s-01", year, month_str))
      if (month_start > Sys.Date()) next
      first_day <- format(month_start, "%Y%m%d")
      last_day <- format(month_start %m+% months(1) - days(1), "%Y%m%d")

      for (sat in satellites) {
        g <- sub("goes", "g", sat)
        urls <- c(urls, sprintf(url_pattern, year, month_str, sat, g,
                                first_day, last_day))
      }
    }
  }
  urls
}

# GOES-R (16/18) science-quality archive: one NetCDF file per satellite/year, e.g.
# .../goes16/l2/data/xrsf-l2-avg1m_science/sci_xrsf-l2-avg1m_g16_y2018_v2-2-1.nc
goes_r_xrs_urls <- function(url_pattern,
                            years = 2017:as.integer(format(Sys.Date(), "%Y")),
                            satellites = c("goes16", "goes18")) {

  urls <- character(0)
  for (sat_dir in satellites) {
    satellite <- sub("goes", "g", sat_dir)
    for (year in years) {

```

```

    urls <- c(urls, sprintf(url_pattern, sat_dir, satellite, year))
  }
}
urls
}

```

This will work for the “NOAA solar flare event lists”, across both the legacy CSV archive and the GOES-R NetCDF archive. Be advised, that the data is downloaded to “data” folder below the data root. This is in order to keep the root folder clean. This data set is about 1.3GB, so we’ll add a file for determining whether or not to download the actual data. In order to run the data download again (takes about 1 hr), delete the “noaa.lock” file in the data folder.

```

# Download every legacy GOES XRS 1-minute CSV file, and every GOES-R
# science-quality NetCDF file, into ~/data - preserving each archive's own
# folder structure from the server
solar_flare_noaa_legacy_xrs_url_pattern <- paste0(
  "https://www.ncei.noaa.gov/data/goes-space-environment-monitor/",
  "access/avg/%d/%s/%s/csv/%s_xrs_1m_%s_%s.csv")
noaa_goes_r_url_pattern <- paste0(
  "https://data.ngdc.noaa.gov/platforms/solar-space-observing-satellites/",
  "goes/%s/l2/data/xrsf-l2-avg1m_science/sci_xrsf-l2-avg1m_%s_y%d_v2-2-1.nc")

# "Lock file" preventing further download
lock_file <- "data/noaa.lock"

# Check for presence of lock file.
# If present, don't download data.
# If not present, download data
if (!file.exists(lock_file)){
  download_files(goes_legacy_xrs_urls(solar_flare_noaa_legacy_xrs_url_pattern), destination_folder = "data")
  download_files(goes_r_xrs_urls(noaa_goes_r_url_pattern), destination_folder = "data")

  # Create file to prevent download
  cat("Done", file = lock_file)
}

```

Last thing to handle is using the Query API towards the Earthquake database as described in Earthquake API - <https://earthquake.usgs.gov/fdsnws/event/1/> .

From versions downloads, we only have data from 1995 until 2020, so we start by limiting our scope to 1995-2020. And since earthquakes below 2.0 will not be felt, we limit the minimum size to 2.5.

The API caps the queries in chunks of 20,000 rows, so we need a loop for this.

```

# Query the USGS Earthquake Catalog and save the response as a CSV file,
# reusing download_file() so a month already on disk is skipped.
# This logic could be optimized, so the check is performed prior to querying
# but this is outside the scope.
#
# The API caps a single request at 20,000 rows, so the date range is
# paginated by month, with each month saved to its own file named after
# the first day of that month.
start_date = lubridate::date("1995-01-01")
end_date = Sys.Date()
# We're also not interested in earthquakes less than 2.5
min_magnitude = 2.5

# We create a function downloading data in this interval saving files in ugs_<date>.csv files
download_usgs_quakes <- function(start_date, end_date, min_magnitude = 2.5,
  destination_folder = "data/ugs") {

  # Create a sequence of month

```

```

months <- seq(as.Date(start_date), as.Date(end_date), by = "month")

# Iterate through the month and download the query result as a file
for (month_start in months) {
  month_start <- as.Date(month_start, origin = "1970-01-01")
  month_end <- month_start %m+% months(1) - days(1)

  # Generate the URL for the Query
  url <- paste0("https://earthquake.usgs.gov/fdsnws/event/1/query?format=csv&",
    sprintf("starttime=%s&endtime=%s&minmagnitude=%s&limit=20000",
      format(month_start, "%Y-%m-%d"), format(month_end, "%Y-%m-%d"),
      min_magnitude)
    )

  # Generate the destination file
  destination_file <- file.path(destination_folder, sprintf("ugs_%s.csv", format(month_start, "%Y-%m-%d")))

  # Attempt download and ignore missing files/errors
  tryCatch(download_file(url, destination_file), error = function(e) NULL)
}
}

```

This will work for the “USGS Earthquake Catalog”, saving one file per month into “data/ugs”. Downloading this data takes up about 164MB of space. So, again we will add a “lock” file, “ugs.lock” in the data folder. Delete this file to download again.

```

# Download every month of earthquakes (magnitude >= 2.5) into data/ugs
usgs_start_date <- "1975-01-01"
usgs_end_date <- Sys.Date()

# "Lock file" preventing further download
lock_file <- "data/ugs.lock"

# Check for presence of lock file.
# If present, don't download data.
# If not present, download data
if (!file.exists(lock_file)){
  # Run the actual download
  download_usgs_quakes(usgs_start_date, usgs_end_date)

  # Create file to prevent download
  cat("Done", file = lock_file)
}

```

Data Analysis

Having downloaded all files (approximately 1.88GB), let’s start looking into the various files. We’ll start by having a look at the raw data file by file.

Parsing the data sources

geomag_kp_gfz.txt

This file holds data dating back to January 1st 1932 measuring sunspots and geomagnetic information. It is a text file with comments identified by #. An extract of this is (modified to fix page):

```
# The parameters in each line are:
#YYY MM DD days days_m Bsr dB Kp1 Kp2 Kp3 Kp4 Kp5 Kp6 Kp7
Kp8 ap1 ap2 ap3 ap4 ap5 ap6 ap7 ap8 Ap SN F10.7obs F10.7adj D
1932 01 01 0 0.5 1352 10 3.333 2.667 2.333 2.667 3.333 2.667 3.333
3.333 18 12 9 12 18 12 18 18 15 22 -1.0 -1.0 2
```

From this, we see, that there are 28 columns in the row. Taking inspiration from the header of the file itself, it seems, that we can limit in information, we need: Kp and ap are three-hourly values. However, the Ap is the mean value of the planetary amplitude (Kp are derived from ap). SN is the “International Solar Spot Number”. F10.7x are the 10.7 cm Solar Flux. So, we will limit the data to date (year/month/day), Ap (mean), SN and F10.7adj.

Processing the file requires basically 3 steps:

- Reading the raw file
- Removing the comments
- Parsing each data line into a table while filtering

```
# Read raw data lines from file
geomag_kp_raw <- readLines(geomag_kp_gfz_data_file)
# Filter out the lines starting with '#'
geomag_kp_raw <- geomag_kp_raw[!startsWith(geomag_kp_raw, "#")]

# 28 columns confirmed against the file's own header/format string:
# "YYY MM DD days days_m Bsr dB Kp1..Kp8 ap1..ap8 Ap SN F10.7obs F10.7adj D"
geomag_data <- read_table(
  # Input data from filtered lines
  I(geomag_kp_raw),
  # Setup column names
  col_names = c(
    "year", "month", "day", "days", "days_m", "bsr", "dB",
    paste0("Kp", 1:8), paste0("ap", 1:8),
    "Ap", "SN", "F107obs", "F107adj", "definitive"
  ),
  col_types = cols(.default = "d")
) %>%
  # Remove all pre 1970 data - we only have data from 1975 elsewhere.
  filter(year >= 1970) %>%
  # Only use subset of columns
  select(year, month, day, Ap, SN, F10.7Adj = F107adj)

# Display as a formatted table
knitr::kable(
  head(geomag_data),
  caption = "Geomagnetic data"
)
```

Table 1: Geomagnetic data

year	month	day	Ap	SN	F10.7Adj
1970	1	1	8	163	150.8
1970	1	2	30	118	144.5
1970	1	3	9	110	137.1
1970	1	4	3	98	131.5
1970	1	5	6	93	126.5
1970	1	6	5	81	117.6

```
cat(format(nrow(geomag_data), big.mark = ","), " rows of Geomagnetic data\n")
```

20,670 rows of Geomagnetic data

Please note, that instead of using the 1975 (which is the limit from other data sources used), we use the 1970: The Earth is a very large system, so the time between an external influence to a reaction might be counted in years. We will leave the data in the geomag_data table for now, and move on to the sunspot_silo.

sunspot_silso.csv

The sunspot_silso file is a basic comma separated file without any headers:

```
1818;01;01;1818.001; -1; -1.0; 0;1
```

However, when you visit Info - <https://www.sidc.be/SILSO/infosndtot>, we'll get the description of the file:

CSV

Filename: SN_d_tot_V2.0.csv

Format: Comma Separated values (adapted for import in spreadsheets)

The separator is the semicolon ';'.

Contents:

Column 1-3: Gregorian calendar date

- Year

- Month

- Day

Column 4: Date in fraction of year.

Column 5: Daily total sunspot number. A value of -1 indicates that no number is available for that day (missing value).

Column 6: Daily standard deviation of the input sunspot numbers from individual stations.

Column 7: Number of observations used to compute the daily value.

Column 8: Definitive/provisional indicator. '1' indicates that the value is definitive. '0' indicates that the value is still provisional.

Looking at this info, we start by seeing, that they use the Gregorian calendar (introduced by Pope Gregory XIII in 1582), which is good, because that's the most used calendar in the world. No conversion is necessary. Based on this info, we can see, that we need year, month, day and total number of sunspots filtering out missing numbers and provisional numbers.

```
# Read raw data lines from file
sunspots_raw <- readLines(sunspot_silso_data_file)

# 8 columns confirmed against the file's info file.
# Data is delimited by ;
sunspots_data <- read_delim(
  # Input data from the rawlines
  I(sunspots_raw),
  # Delimiter is ;
  delim = ";",
  # Setup column names
  col_names = c(
    "year", "month", "day", "dateFraction", "total", "stdDev", "noOfStations",
    "definitive"),
  col_types = cols(.default = "d"),
  # Some of the columns are padded with whitespaces, so these need to be removed
  trim_ws = TRUE
) %>%
  # Remove all pre 1970 data - we only have data from 1975 elsewhere.
  filter(year >= 1970 & definitive > 0) %>%
```

```

select (year, month, day, total)

# Display as a formatted table
knitr::kable(
  head(sunspots_data),
  caption = "Sunspot data"
)

```

Table 2: Sunspot data

year	month	day	total
1970	1	1	163
1970	1	2	118
1970	1	3	110
1970	1	4	98
1970	1	5	93
1970	1	6	81

```
cat(format(nrow(sunspots_data), big.mark = ","), " rows of Sunspot data\n")
```

20,544 rows of Sunspot data

Again, please note, that instead of using the 1975 (which is the limit from other data sources used), we use the 1970: The Earth is a very large system, so the time between an external influence to a reaction might be counted in years. In case of this “delay” being counted in years, one discuss whether or not, this “warning” would be relevant (on one hand, an early warning allows for preparation, on the other, political interests may not take this warning into account).

`ugs_[year]_[month]_[day].csv`

The UGS data are a series of .csv files containing the information about the Earthquakes:

Column	Description	Included?
time	Time stamp of the logged event	Yes
latitude	Latitude of the center of the event	N/A
longitude	Longitude of the center of the event	N/A
depth	Depth of the event	N/A
mag	Magnitude of the event	Yes
magType	Magnitudetype used for internal filtering	N/A
nst	Number of seismic stations used to determine location	N/A
gap	Largest azimuth gap between adjacent stations	N/A
dmin	Horizontal distance from epicenter to nearest station	N/A
rms	RMS value of the travel time	N/A
net	ID of the preferred data contributor	N/A
id	Unique ID	N/A
updated	Last time of update	N/A
place	Geographical name of the epicenter	N/A
type	Type of the seismic event	Yes, filtering
horizontalError	Uncertainty of distance the epicenter	N/A
depthError	Uncertainty of depth the epicenter	N/A
magError	Uncertainty of magnitude the epicenter	N/A
magNst	Total number of seismic station reporting the event	N/A
status	Status of the event	Yes, filtering
locationSource	Original reporter of the location this event	N/A
magSource	Original reporter of the magnitude this event	N/A

This clearly shows, that we only need 4 columns (time, magnitude for further processing and type, status for filtering). For filtering, we’re only interested in type = “earthquake”, since we do not care about man made events such as

quarrying for stone. The status field itself can hold 3 values: “automatic”, “reviewed”, “deleted”. For this analysis, I’ll treat “automatic” and “reviewed” the same - whether or not the staff hasn’t reviewed it yet, so these will be included. “deleted” will not be used.

Now, we already have the files underneath the data/ugs folder, so we need to traverse all 621 (at the time of downloading the data) files and add all of the data to a filtered table:

```
# Force the current position to be the folder of the current file
knitr::opts_knit$set(root.dir = dirname(knitr::current_input(dir = TRUE)))
# Get the list of csv files in folder
ugs_files <- list.files("data/ugs", pattern = "\\..csv$", full.names = TRUE)

# Read only the columns we need from every file, then combine into a single table
ugs_data <- map_dfr(
  ugs_files,
  ~ read_csv(
    .x,
    col_types = cols_only(
      time = col_datetime(format = ""),
      mag = col_double(),
      type = col_character(),
      status = col_character()
    )
  )
) %>%
  # Only keep actual earthquakes, and drop "deleted" entries
  filter(type == "earthquake" & status != "deleted")

# Display as a formatted table
knitr::kable(
  head(ugs_data),
  caption = "UGS data"
)
```

Table 3: UGS data

time	mag	type	status
1975-01-30 20:11:19	4.20	earthquake	reviewed
1975-01-30 19:02:39	2.92	earthquake	reviewed
1975-01-30 16:26:21	3.40	earthquake	reviewed
1975-01-30 15:55:10	2.63	earthquake	reviewed
1975-01-30 14:48:40	3.70	earthquake	reviewed
1975-01-30 14:03:18	3.13	earthquake	reviewed

```
cat(format(nrow(ugs_data), big.mark = ","), " rows of UGS data\n")
```

973,419 rows of UGS data

NOAA data

The NOAA data is a bit more challenging from a data wrangling point of view: The data format changes by 2020 3rd month from a text format to something resembling a binary format (NetCDF - Network Common Data Form). Luckily, R has support for parsing these binary files.

The text format is a combination of multiple csv files with a description “header” inside a single file. Digging deeper into the data, it seems that the newer satellites renamed the channels from xs/xl to A_AVG/B_AVG. So, we need to compensate for this:

```

# Get the list of csv files in subfolders
noaa_files1 <- list.files("data/data/goes-space-environment-monitor/access/avg",
                        pattern = "\\\\.csv$", full.names = TRUE, recursive = TRUE)

# Read only the columns we need from every file, then combine into a single table
# Take into account, that newer satellite data has changed format:
# goes08-goes12 : time, xs, xl
# goes13-goes15 : time, A_AVG, B_AVG

# Create function for parsing goes file.
read_goes_file <- function(path) {
  # Read lines from file
  raw <- readLines(path)

  # Skip anything preceeding the "data:" marker
  data_start <- which(trimws(raw) == "data:") + 1
  # Extract the header from the data block.
  header <- raw[data_start]
  # Extract the actual data
  body <- raw[data_start:length(raw)]

  # Old satellites use the "time_tag,xs,xl" tags, so we know, that this is the
  # old satellite data.
  if (startsWith(header, "time_tag,xs,xl")) {

    # Data columns are already named xs/xl
    parsed <- read_csv(
      # Read data as-is
      I(body),
      col_types = cols(
        # Extract time in the specific format
        time_tag = col_datetime(format = "%Y-%m-%d %H:%M:%OS"),
        xs = col_double(),
        xl = col_double()
      ),
      na = "-99999.0"
    )
  } else {
    # Newer satellites: A_AVG/B_AVG replace xs/xl, plus quality-flag columns we don't need
    parsed <- read_csv(
      # Read data as-is
      I(body),
      col_types = cols_only(
        # Extract time in the specific format
        time_tag = col_datetime(format = "%Y-%m-%d %H:%M:%OS"),
        A_AVG = col_double(),
        B_AVG = col_double()
      ),
      na = "-99999.0"
    ) %>%
    # Rename the new columns to match the old format
    rename(xs = A_AVG, xl = B_AVG)
  }

  # Common for both formats
  parsed %>%
  # Rename time stamp column
  rename(time = time_tag) %>%
  # Return data
  select(time, xs, xl) %>%

```

```

  filter(!is.na(xs) & !is.na(xl))
}

# Parse every file (674 total) and combine into a single table
goes_data <- map_dfr(noaa_files1, read_goes_file)

# Display as a formatted table
knitr::kable(
  head(goes_data),
  caption = "NOAA data part 1"
)

```

Table 4: NOAA data part 1

time	xs	xl
1995-01-03 00:00:00	0	1e-07
1995-01-03 00:01:00	0	1e-07
1995-01-03 00:02:00	0	1e-07
1995-01-03 00:03:00	0	1e-07
1995-01-03 00:04:00	0	1e-07
1995-01-03 00:05:00	0	1e-07

```
cat(format(nrow(goes_data), big.mark = ","), " rows of NOAA data in the first location\n")
```

22,575,435 rows of NOAA data in the first location

From 2020 onwards (GOES-16 and GOES-18), NOAA switched to publishing the data as NetCDF files instead of the header/csv format above. The channel columns are renamed again, this time to `xrsa_flux` (previously `xs/A_AVG`) and `xrsb_flux` (previously `xl/B_AVG`), missing values are flagged with `-9999` instead of `-99999.0`. The timestamp is stored as seconds since 2000-01-01 12:00:00 UTC rather than a formatted date string. We parse these the same way and append the result onto the existing `goes_data` table:

```

# Get the list of NetCDF files in subfolders
noaa_files2 <- list.files("data/platforms/solar-space-observing-satellites/goes",
  pattern = "\\\\.nc$", full.names = TRUE, recursive = TRUE)

# Create function for parsing a single GOES NetCDF file
read_goes_nc_file <- function(path) {
  # Open the file, ensuring it is always closed again afterwards
  nc <- nc_open(path)
  on.exit(nc_close(nc), add = TRUE)

  # Time is stored as seconds since 2000-01-01 12:00:00 UTC (per the file's own "units" attribute)
  time_secs <- ncvar_get(nc, "time")
  time <- ymd_hms("2000-01-01 12:00:00", tz = "UTC") + seconds(time_secs)

  # xrsa_flux/xrsb_flux are the short/long wavelength channels - same as xs/xl above
  xs <- ncvar_get(nc, "xrsa_flux")
  xl <- ncvar_get(nc, "xrsb_flux")

  tibble(time = time, xs = xs, xl = xl) %>%
    # -9999 is this dataset's missing-value sentinel (its own _FillValue, not -99999.0)
    mutate(xs = na_if(xs, -9999), xl = na_if(xl, -9999)) %>%
    filter(!is.na(xs) & !is.na(xl))
}

# Parse every NetCDF file and append the result onto the existing goes_data table
goes_data <- bind_rows(goes_data, map_dfr(noaa_files2, read_goes_nc_file))

```

```
# Display as a formatted table
knitr::kable(
  head(goes_data),
  caption = "NOAA data part 2"
)
```

Table 5: NOAA data part 2

time	xs	xl
1995-01-03 00:00:00	1.11e-09	9.73e-08
1995-01-03 00:01:00	1.36e-09	9.57e-08
1995-01-03 00:02:00	1.23e-09	9.46e-08
1995-01-03 00:03:00	1.21e-09	1.01e-07
1995-01-03 00:04:00	1.43e-09	1.05e-07
1995-01-03 00:05:00	1.66e-09	1.22e-07

```
cat(format(nrow(goes_data), big.mark = ","), " rows of NOAA data after adding the second location\n")
```

28,906,984 rows of NOAA data after adding the second location

Visualization of the data

Now, that we have the data in the `geomag_data`, `sunspots_data`, `ugs_data` and `goes_data` tables, we can start taking a look at the data in more details. Let's start visualizing the data on a time line. First, we'll start by setting up a few conditions:

1. Due to space constraints, I'll will only show data based on the month, not the daily/hourly/minute/second information.
2. We'll set up a function for plotting the data, so we don't need to repeat code.

```
# Earthquakes: average magnitude per month
eq_monthly <- ugs_data %>%
  mutate(month = floor_date(time, "month")) %>%
  group_by(month) %>%
  summarise(avg_mag = mean(mag, na.rm = TRUE), .groups = "drop")

# Geomagnetic Ap index: monthly mean
geomag_monthly <- geomag_data %>%
  mutate(month = floor_date(make_date(year, month, day), "month")) %>%
  group_by(month) %>%
  summarise(avg_Ap = mean(Ap, na.rm = TRUE), .groups = "drop")

# Sunspot number: monthly mean (-1 marks a missing daily reading)
sunspot_monthly <- sunspots_data %>%
  mutate(total = na_if(total, -1)) %>%
  mutate(month = floor_date(make_date(year, month, day), "month")) %>%
  group_by(month) %>%
  summarise(avg_sunspots = mean(total, na.rm = TRUE), .groups = "drop")

# GOES X-ray flux (xl channel): monthly mean
goes_monthly <- goes_data %>%
  mutate(month = floor_date(time, "month")) %>%
  group_by(month) %>%
  summarise(avg_xl = mean(xl, na.rm = TRUE), .groups = "drop")

# Create a generic function for displaying earthquakes in relation to other data
```

```

plot_dual_axis_earthquakes <- function(secondary, secondary_col,
                                       secondary_label,
                                       title) {

  # Rescale the secondary series into the earthquake series' range so
  # both draw together
  primary_range <- range(eq_monthly$avg_mag, na.rm = TRUE)
  secondary_range <- range(secondary[[secondary_col]], na.rm = TRUE)
  scale_factor <- diff(primary_range) / diff(secondary_range)
  offset <- primary_range[1] - secondary_range[1] * scale_factor

  secondary <- secondary %>%
    mutate(scaled = .data[[secondary_col]] * scale_factor + offset)

  # Only show data inside the 1975-2026 range
  plot_start <- as.Date("1975-01-01")
  plot_end <- as.Date("2026-12-31")
  eq_plot_data <- eq_monthly %>% filter(month >= plot_start & month <= plot_end)
  secondary <- secondary %>% filter(month >= plot_start & month <= plot_end)

  # Plot the actual data
  ggplot() +
    geom_line(data = eq_plot_data, aes(x = month, y = avg_mag, color = "Avg. earthquake magnitude"),
              linewidth = 0.7) +
    geom_line(data = secondary, aes(x = month, y = scaled, color = secondary_label),
              linewidth = 0.7) +
    # Map the two series to explicit colors so a legend box is drawn for each
    scale_color_manual(
      name = NULL,
      values = setNames(c("steelblue", "firebrick"), c("Avg. earthquake magnitude", secondary_label))
    ) +
    scale_y_continuous(
      name = "Avg. earthquake magnitude",
      sec.axis = sec_axis(~ (. - offset) / scale_factor, name = secondary_label)
    ) +
    scale_x_date(date_breaks = "1 year", date_labels = "%Y",
                 limits = c(plot_start, plot_end)) +
    # Zoom the view to start at magnitude 3 (we don't record anything below that)
    # without dropping any underlying data
    coord_cartesian(ylim = c(3, NA)) +
    labs(title = title, x = "Year") +
    theme_minimal() +
    theme(
      axis.text.x = element_text(angle = 90, hjust = 1),
      axis.text.y = element_text(size = 8),
      axis.text.y.right = element_text(size = 8),
      panel.grid.minor.x = element_blank(), # no sub-year gridlines
      legend.position = "bottom"
    )
}

```

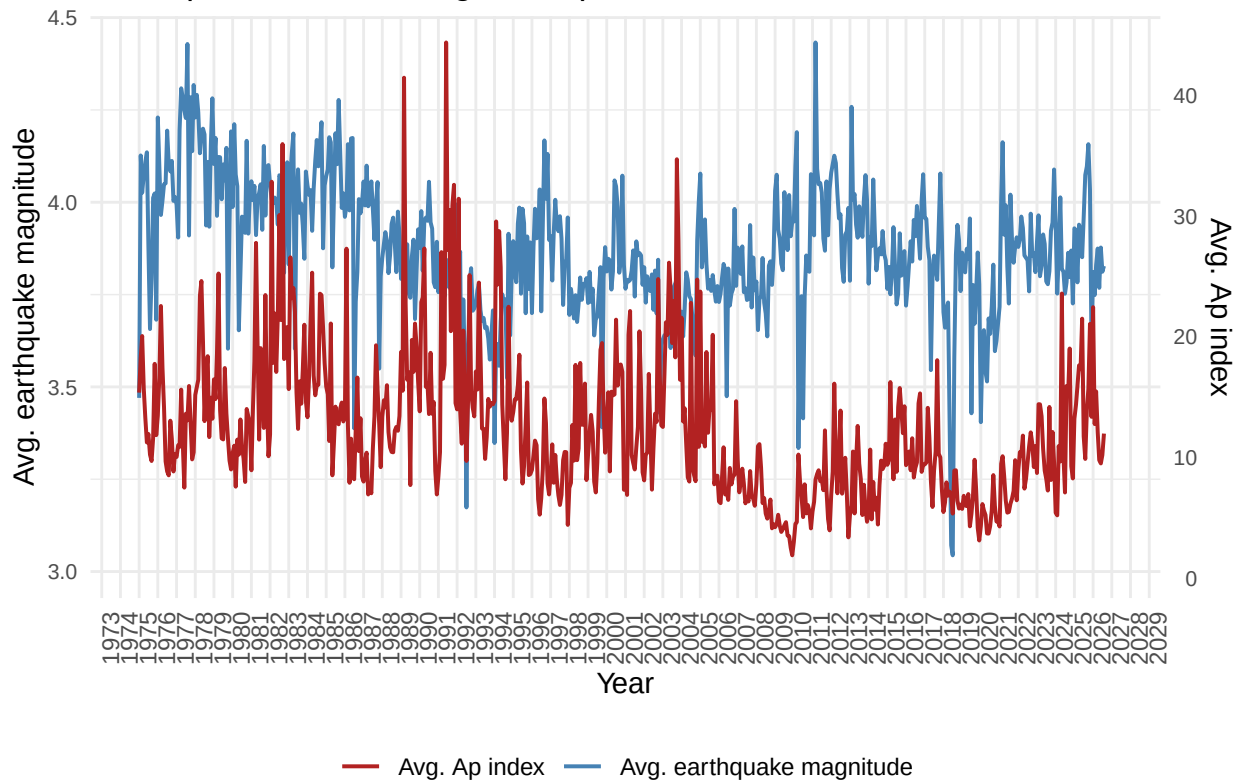
Earthquakes vs geomagnetism

```

# Graph 1: Earthquakes vs Geomagnetic Ap index
plot_dual_axis_earthquakes(geomag_monthly, "avg_Ap", "Avg. Ap index",
                           "Earthquakes vs Geomagnetic Ap Index")

```

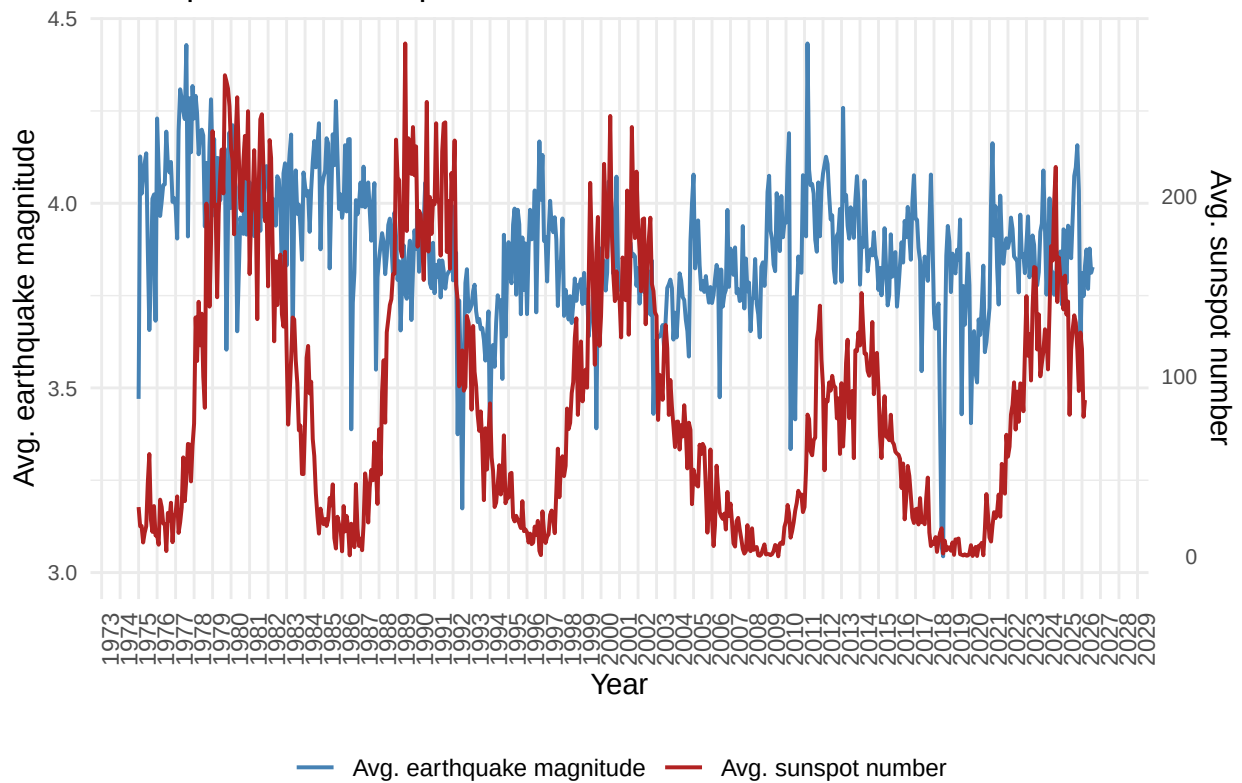
Earthquakes vs Geomagnetic Ap Index



Graph 2: Earthquakes vs Sunspot number

```
plot_dual_axis_earthquakes(sunspot_monthly, "avg_sunspots", "Avg. sunspot number",
                           "Earthquakes vs Sunspot Number")
```

Earthquakes vs Sunspot Number



```
# Graph 3: Earthquakes vs GOES X-ray flux
```

```
plot_dual_axis_earthquakes(goes_monthly, "avg_xl", "Avg. X-ray flux (xl, W/m²)",  
                           "Earthquakes vs GOES X-ray Flux")
```

